

Quantum Mechanics in Solids

Unit-5

Lecture-4

Bloch's theorem for particles in a periodic potential

Bloch Theorem

Variable Potential $\rightarrow$ Periodic Potential

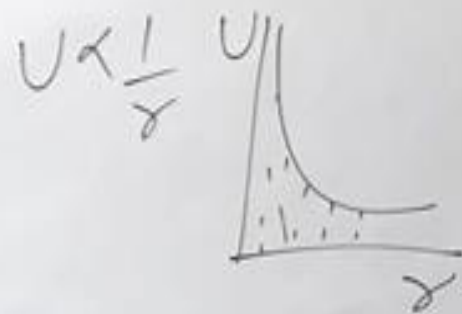
$$\psi = ?$$

Metal



Constant or Zero Potential

Bloch Theorem

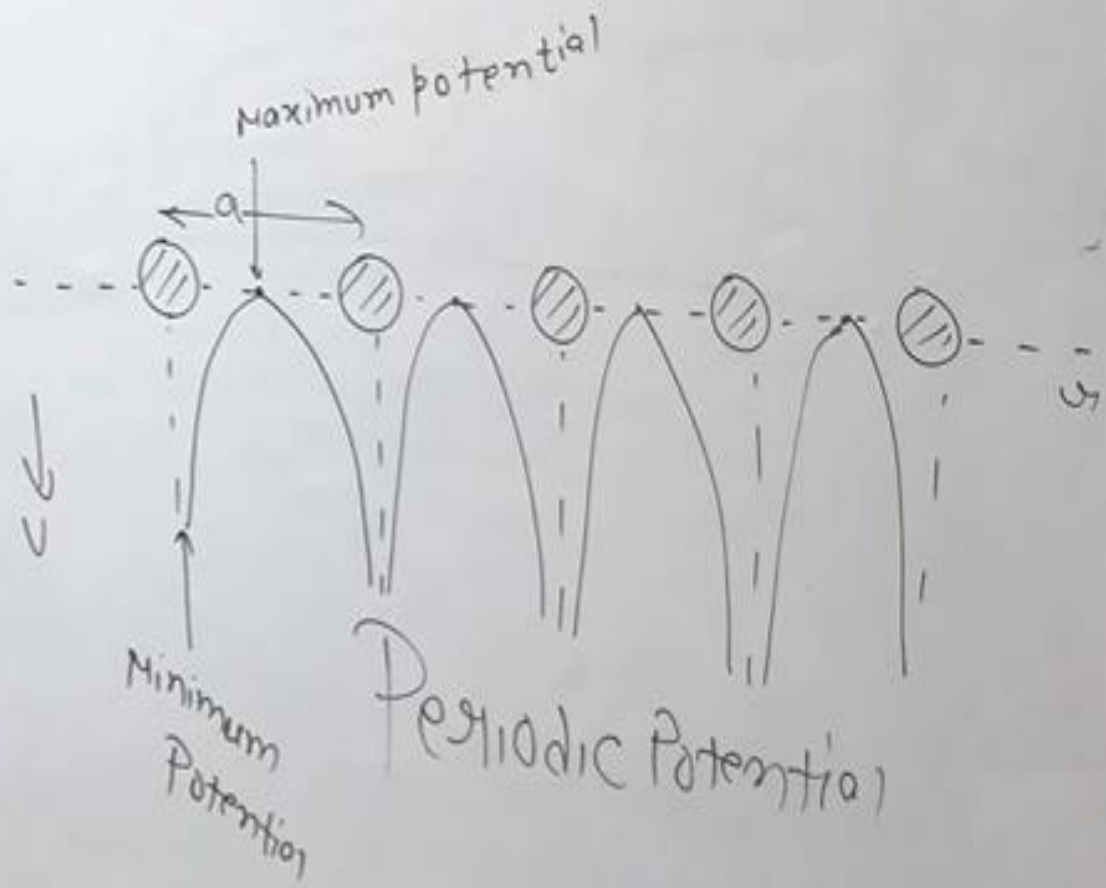


$$U = \frac{1}{4\pi\epsilon_0} \times \frac{ze(-e)}{r}$$

$$\downarrow U = -\frac{1}{4\pi\epsilon_0} \times \frac{ze^2}{r} \uparrow \downarrow$$

$$\downarrow U \propto \frac{1}{r} \uparrow$$

$$V(x+a) = V$$



Bloch Theorem

Statement of Bloch Theorem:

$$\frac{d^2 \psi}{dx^2} + \frac{2m}{\hbar^2} [E - V(x)] \psi = 0 \quad (1)$$

If electron is in Periodic Potential
 $V(x+a) = V$

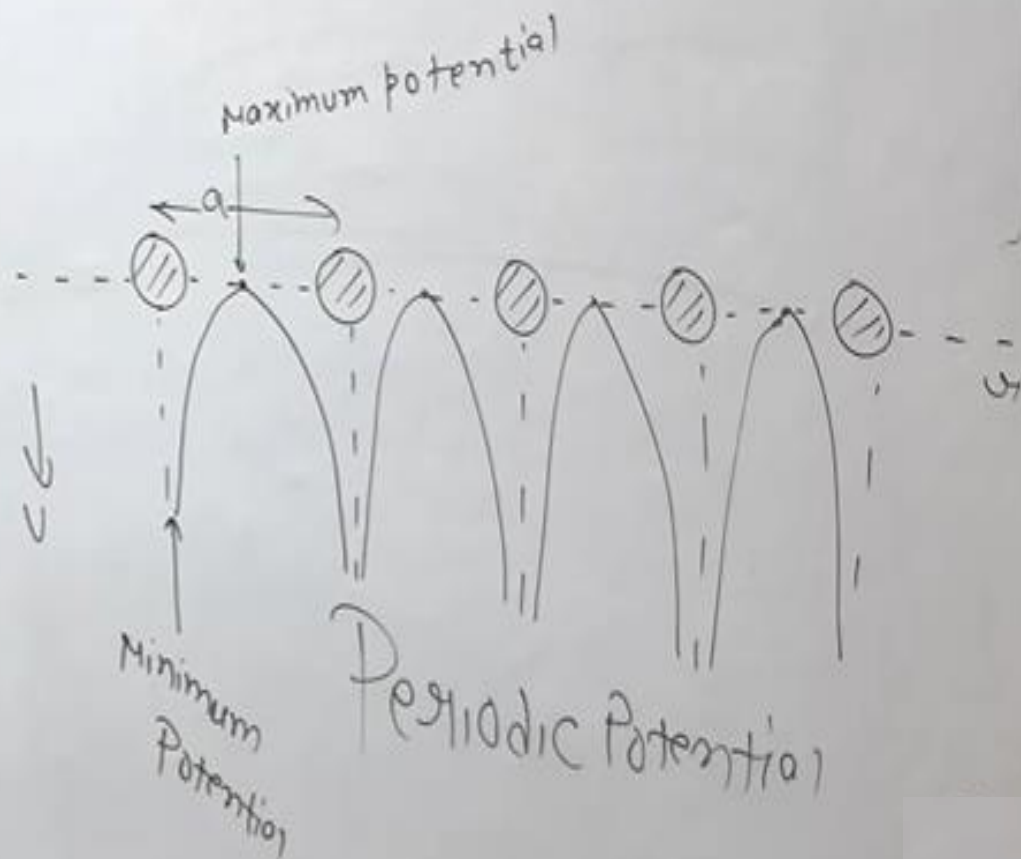
$$\psi(x) = e^{\pm i k x} u_k(x)$$

Bloch Function

$$u_k(x+a) = u_k(x)$$

$$\psi(x+a) = e^{\pm i k x} e^{\pm i k a} u_k(x+a)$$

$$\psi(x+a) = e^{\pm i k a} \psi(x)$$



Proof of Bloch Theorem:

Schrodinger Eqⁿ of Electron
moving in 1-D periodic potential

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} [E - V(x)] \psi = 0 \quad \text{--- (1)}$$

$$V(x+a) = V(x)$$

Let
 $f(x)$ and $g(x)$ are two independent
solⁿ of above Eqⁿ

$$\psi(x) = Af(x) + Bg(x) \quad \text{--- (2)}$$

$f(x+a)$ & $g(x+a)$ independent solutions

$$\left. \begin{aligned} f(x+a) &= \alpha_1 f(x) + \alpha_2 g(x) \\ g(x+a) &= \beta_1 f(x) + \beta_2 g(x) \end{aligned} \right\} \quad \text{--- (3)}$$

$$\psi(x+a) = Af(x+a) + Bg(x+a)$$

$$\psi(x+a) = A[\alpha_1 f(x) + \alpha_2 g(x)] + B[\beta_1 f(x) + \beta_2 g(x)]$$

$$\psi(x+a) = (A\alpha_1 + B\beta_1)f(x) + (A\alpha_2 + B\beta_2)g(x)$$

Choose

$$\left. \begin{aligned} A\alpha_1 + B\beta_1 &= Q A \\ A\alpha_2 + B\beta_2 &= Q B \end{aligned} \right\} \quad \text{--- (4)}$$

$$\psi(x+a) = Q [Af(x) + Bg(x)]$$

$$\psi(x+a) = Q \psi(x) \quad \text{--- (5)}$$

From Eqⁿ (4)

$$(\alpha_1 - Q)A + (\beta_1)B = 0$$

$$(\alpha_2)A + (\beta_2 - Q)B = 0$$

$$\begin{vmatrix} \alpha_1 - Q & \beta_1 \\ \alpha_2 & \beta_2 - Q \end{vmatrix} = 0$$

$$Q^2 - (\alpha_1 + \beta_2)Q + \alpha_1\beta_2 - \alpha_2\beta_1 = 0 \quad \text{--- (6)}$$

$$\therefore \alpha_1\beta_2 - \alpha_2\beta_1 = 1$$

$f(x)$ and $g(x)$ are solutions

$$\frac{d^2 f(x)}{dx^2} + \frac{2m}{\hbar^2} [E - V(x)] f(x) = 0 \quad \text{--- (7)}$$

$$\frac{d^2 g(x)}{dx^2} + \frac{2m}{\hbar^2} [E - V(x)] g(x) = 0 \quad \text{--- (8)}$$

Multiply Eqⁿ (7) by $g(x)$

Eqⁿ (8) by $f(x)$, subtract

$$g(x) \frac{d^2 f(x)}{dx^2} - f(x) \frac{d^2 g(x)}{dx^2} = 0$$

Integrate

$$\underbrace{g(x) \frac{df(x)}{dx} - f(x) \frac{dg(x)}{dx}}_{\text{Wronskian } W(x)} = \text{constant}$$

$$W(x) = g(x) \frac{df(x)}{dx} - f(x) \frac{dg(x)}{dx} \quad \text{--- (9)}$$

$$W(x+a) = W(x)$$

$$W(x+a) = g(x+a) \frac{df(x+a)}{dx} - f(x+a) \frac{dg(x+a)}{dx}$$

$$W(x+a) = \left[\beta_1 f(x) + \beta_2 g(x) \right] \frac{d \left[\alpha_1 f(x) + \alpha_2 g(x) \right]}{dx} - \left[\alpha_1 f(x) + \alpha_2 g(x) \right] \frac{d \left[\beta_1 f(x) + \beta_2 g(x) \right]}{dx}$$

$$\begin{vmatrix} \alpha_1 - Q & \beta_1 \\ \alpha_2 & \beta_2 - Q \end{vmatrix} = 0$$

$$Q^2 - (\alpha_1 + \beta_2)Q + \alpha_1\beta_2 - \alpha_2\beta_1 = 0 \quad \text{--- (6)}$$

$$\therefore \alpha_1\beta_2 - \alpha_2\beta_1 = 1$$

$f(x)$ and $g(x)$ are solutions

$$\frac{d^2 f(x)}{dx^2} + \frac{2m}{\hbar^2} [E - V(x)] f(x) = 0 \quad \text{--- (7)}$$

$$\frac{d^2 g(x)}{dx^2} + \frac{2m}{\hbar^2} [E - V(x)] g(x) = 0 \quad \text{--- (8)}$$

Multiply Eqⁿ (7) by $g(x)$

Eqⁿ (8) by $f(x)$, subtract

$$g(x) \frac{d^2 f(x)}{dx^2} - f(x) \frac{d^2 g(x)}{dx^2} = 0$$

Integrate

$$\underbrace{g(x) \frac{df(x)}{dx} - f(x) \frac{dg(x)}{dx}}_{\text{Wronskian } W(x)} = \text{constant}$$

$$W(x) = g(x) \frac{df(x)}{dx} - f(x) \frac{dg(x)}{dx} \quad \text{--- (9)}$$

$$W(x+a) = W(x)$$

$$W(x+a) = g(x+a) \frac{df(x+a)}{dx} - f(x+a) \frac{dg(x+a)}{dx}$$

$$W(x+a) = \left[\beta_1 f(x) + \beta_2 g(x) \right] \frac{d \left[\alpha_1 f(x) + \alpha_2 g(x) \right]}{dx} - \left[\alpha_1 f(x) + \alpha_2 g(x) \right] \frac{d \left[\beta_1 f(x) + \beta_2 g(x) \right]}{dx}$$

$$w(x+a) = \alpha_1 \beta_2 - \alpha_2 \beta_1 \left[g(x) \frac{df(x)}{dx} - f(x) \frac{dg(x)}{dx} \right]$$

$$w(x+a) = \alpha_1 \beta_2 - \alpha_2 \beta_1 \quad w(x)$$

$$\alpha_1 \beta_2 - \alpha_2 \beta_1 = 1$$

So, Eqⁿ (6) becomes

$$Q^2 - (\alpha_1 + \beta_2)Q + 1 = 0$$

$$Q^2 - \mu Q + 1 = 0 \quad (10)$$

CASE-1 $\mu^2 < 4$

Q_1 and Q_2 are complex
Conjugate Roots

$$Q_1 = e^{ika}$$

$$Q_2 = e^{-ika}$$

$$\psi(x+a) = e^{\pm ika} \psi(x)$$

Statement of Bloch Theorem

Allowed
Energy
levels

$$Q^2 - \mu Q + 1 = 0 \quad (10)$$

$$Q_1 = \frac{\mu + \sqrt{\mu^2 - 4}}{2}$$

$$Q_2 = \frac{\mu - \sqrt{\mu^2 - 4}}{2}$$

CASE-2 $\mu^2 > 4$

Q_1 and Q_2 are Real Roots

It means

Absence of wave function

Electron is Absent

Forbidden Gap